Homework 6
Adversarial Attack

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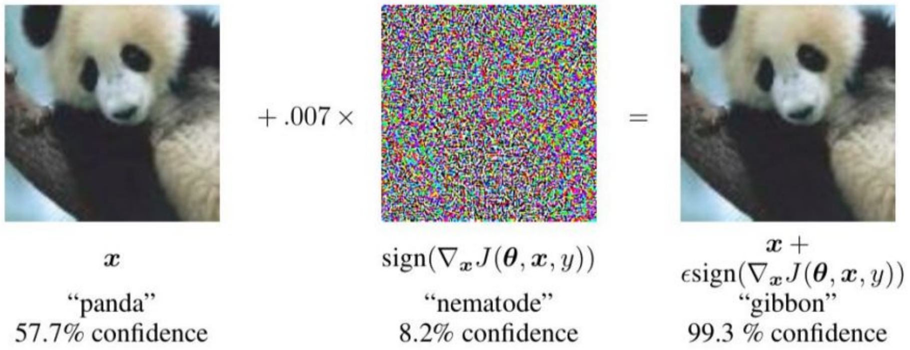
Learning Objective

Task Description - Prerequisite

- Those are methodologies which you should be familiar with first
  - Attack objective: Non-targeted attack
  - Attack constraint: L-infinity norm and Parameter ε (epsilon)
  - Attack algorithm: FGSM attack
  - Attack schema: Black box attack (perform attack on proxy network)
  - Benign images vs Adversarial images



02

Homework Description

Task Description - TODO

1. Fast Gradient Sign Method (FGSM)
    1. Choose any proxy network to attack the black box
    2. Implement non-targeted FGSM from scratch
1. Any methods you like to attack the model
    1. Implement any methods you prefer from scratch
    2. Iterative Fast Gradient Sign Method (I-FGSM) --- medium baseline
    3. Model ensemble attack --- strong/boss baseline

Task Description - FGSM

- Fast Gradient Sign Method (FGSM)

$$x' = x + \epsilon \cdot \text{sign}(\nabla_x L(x, y; \theta))$$

x: benign image

x’: (perturbed) adversarial image

L: classification loss function

y: ground truth of x

Task Description - I-FGSM

- Iterative Fast Gradient Sign Method (I-FGSM)

$$x'_0 = x$$

$$x'_{n+1} = \mathrm{Clip}^{\epsilon}_{x}(x'_n + \alpha \cdot \mathrm{sign}(\nabla_x L(x, y; \theta)))$$

α: step size

n: iteration number

$\mathrm{Clip}^{\epsilon}_{x}$: adversarial images should be within the ϵ-ball (ϵ-limitation) to benign images

Task Description - Ensemble Attack

- Choose a list of proxy models
- Choose an attack algorithm (FGSM, I-FGSM, and so on)
- Attack multiple proxy models at the same time
- Delving into Transferable Adversarial Examples and Black-box Attacks
- Query-Free Adversarial Transfer via Undertrained Surrogates

Task Description - Evaluation Metrics

- Parameter ε is fixed as 8
- Distance measurement: L-inf. norm
- Model Accuracy is the only evaluation metrics

benign

adversarial (\eps = 8)

adversarial (\eps = 16)

Data Format

- Download link: link
- Images:
  - CIFAR-10 images
  - (32 * 32 RGB images) * 200
    - airplane/airplane1.png, …, airplane/airplane20.png
    - …
    - truck/truck1.png, …, truck/truck20.png
  - 10 classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck)
  - 20 images for each class

Data Format

- In this homework, we can perform attack on pretrained models
- Pytorchcv provides multiple models pret-rained on CIFAR-10
- A model list is provided here

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Grading Rubric

Grading Rubric

- 80 points for the coding section
    - Global Settings (10 points)
    - Data (10 points)
    - Utils – Benign Images Evaluation (5 points)
    - Utils – Attack Algorithm (20 points)
    - Utils – Attack (15 points)
    - Model / Loss Function (5 points)
    - FGSM (10 points)
    - Visualization (5 points)
- 20 points for the report

NOTE: Code that cannot be executed successfully will result in a grade of **ZERO**.

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Deadline & Submission

Code & Submission Standard

- **MyCourses Submission**
  - Compress your code and report into

    <student ID>_hw6.zip

    e.g. zh4258_hw6.zip

  - Only latest submission is graded.
  - Do not submit your model or dataset.
  - Make sure code is reasonable and has comments or else grade x 0.9
    - Please find coding standard at: https://peps.python.org/pep-0008/

Code Citation Standard

- You must specify the source of your code.
- E.g., add a text block named **Reference** at the bottom of your code or at the end of the report.

Reference

Source: Goldberg, David E.; Holland, John H. @Genetic Algorithms and Machine Learning (https://deepblue.lib.umich.edu/bitstream/handle/2027.42/46947/10994_2005_Article_422926.pdf;jsessionid=A44A02385EA28F890A4C30D355376C94?sequence=1)

ZIP File Standard

- Your .zip file should include only
  - **Code**: either .py or .ipynb
  - **Report**: .pdf
    - Walk through of your thought process
      - Explanations of your code
    - Citations

Deadline

- **Code Submission (Through MyCourses)**

2026/03/27 23:59 (EST)

No late submission!
No extension!
Submit early!

Regulations Again

- You should finish your homework on your own.
- You should **NOT** modify the generated images manually.
- **Do NOT** share codes or generated images with any living creatures.
- **Do not search or use additional data or pre-trained models.**
- Your **final grade at least x 0.9** if you violate any of the above rules.
- Dr. Liu & TAs preserve the rights to change the rules & grades.